

Combustion

Combustion is the reaction between a fuel—such as wood, natural gas, or oil—and oxygen. The combustion reaction releases energy in the form of heat and light. Fuel needs a trigger (a match or a spark) before combustion can start.

Combustion is at work in bonfires, fireworks, and when we light a candle. But more than just a spectacle, it is essential to the way we live. Most of the world's power stations generate electricity using the combustion of fossil fuels such as coal, oil, and gas. Most cars, semi trucks, boats, and planes are driven by engines powered by combustion. Scientists are working hard to create alternatives to what is now understood to be a potentially wasteful and harmful source of energy. But for now we all rely on it to keep warm and to get where we need.

Campfire chemistry

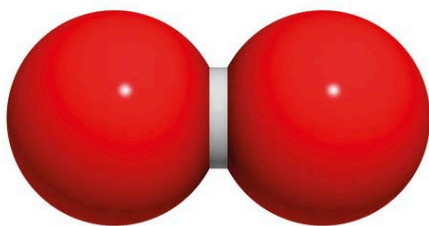
Dry wood contains cellulose (made of the elements carbon, hydrogen, and oxygen). It burns well in oxygen, which makes up about one fifth of air.

Carbon dioxide

Carbon dioxide (CO₂) is produced when wood burns. Known as a greenhouse gas, it contributes to global warming if there is too much of it in the atmosphere.

Water vapor

The combustion of cellulose, which makes up about half the dry mass of wood, produces water (H₂O) as well as carbon dioxide (CO₂). In the heat of a fire, the water evaporates as steam.

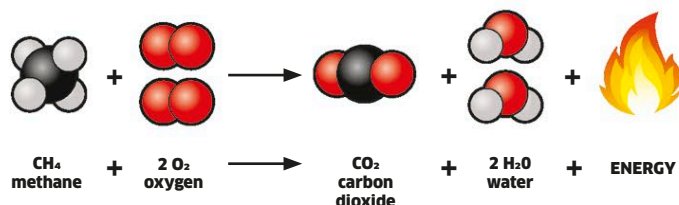


Oxygen

For combustion to work, there needs to be a good supply of the element oxygen. Oxygen in the air exists as molecules made up of two oxygen atoms, with the chemical formula O₂.

A balanced reaction

During combustion, substances known as reactants are transformed into new substances called products. The reaction rearranges the atoms of the reactants. They swap places, but the number of each is the same. Energy (heat and light) is released when the bonds that hold the initial molecules together are broken and new ones are formed.



Methane combustion

Above is the reaction formula for the combustion of methane (natural gas). The number of carbon (C), hydrogen (H), and oxygen (O) atoms is the same on each side of the arrow, but the substances they make up have changed.

